

Observable dictionary: estimator + ensemble + null

Family	Representative estimator	Averaging ensemble	Required control	Allowed reading
Risk	$R_{train}, R_{IID}, R_{OOD}$	evaluation examples	chance / calibrated baseline	prediction quality
Recovery	$\mathbb{I}(m; Z)/H(Z)$	outer seeds + examples	shuffled latent labels	semantic order
Specialization	head / role assignment	permutation-aligned seeds	permuted components	symmetry breaking
Response	$\partial\langle m\rangle/\partial g$	independent outer seeds	window + scale sensitivity	crossover candidate
Fluctuation	$N\text{Var}_Q(m)$	outer disorder only	temporal variance	susceptibility
Dynamics	time-to-order, path entropy	training trajectories	matched compute	accessibility / arrest
Mechanism	$R_{ablated} - R_{full}$	matched interventions	equal-resource sham	causal contribution

Rule: without ensemble, units, validity range, and null, an estimator is not a registered observable.